Technical Report

Dataset description and schema

The dataset used in this project is the TMDB Movies Dataset (2023) containing 1336575 movies, obtained from Kaggle, and each record includes the following fields:

- Movie title
- Vote average
- Status
- Release date
- Revenue
- Runtime
- Budget
- IMDB ID
- Genres

Because the dataset is very large, loading directly from CSV for every program execution would be slow. To solve this problem we implemented two-step system:

CSV Loader

If the pickle file does not exist, the program:

1. Reads the CSV line-by-line
2. Converts each row into a typed Python dictionary
3. Stores it under a unique integer ID
4. Saves the entire dataset as a pickle object

Pickle Loader

If a pickle exists, we skip CSV parsing and directly load the dictionary.

Chosen data structures and justification

AVL trees

To support efficient search and querying, for indexing we use three AVL-balanced binary search trees:

- AVL_title
- AVL_year
- AVL_genre

Operation	Description	Complexity
Search	lookup by title/year/genre	O(log n)
Insert	inserting new movie record + AVL updates	O(log n)
Delete	remove record + fix AVL trees	O(log n)
Modify	update record + possibly reindex affected fields.	O(log n)
Prefix Search	autocompletion-style queries	O(log n + k)

Graphs

- Vertices: Genres (strings)
- Edges**:** Co-occurrence of genres in a movie
- Adjacency Map**:**
- Edge Weights**:** Number of times two genres co-occur in the dataset

Operation	Description	Time Complexity
Insert_vertex	Add a genre vertex if it doesn't exist	O(1)
Insert_edge	Add or increment edge weight between genres	O(1)
Build_from_dataset	Build graph from dataset by connecting co-occurring genres	$O(M * G^2)$
Shortest_path	BFS shortest path between two genres	O(V + E)
Top_k_degrees	Return top-k genres by number of connections	O(V log V)
Genre_popularity_score	Sum popularity of all movies with the given genre	O(M)

Query Engine

Method	Operation Details	Time Complexity
search_by_id(movie_id)	Look up movie in by_id dictionary	**O(1)**
search_by_title(movie_title)	Lookup in AVL (title_idx.get) then dictionary	**O(log N)**
search_by_year(movie_year)	Lookup in AVL (year_idx.get) then fetch records	**O(log N + K)**
search_by_genre(movie_genre)	Lookup in AVL (genre_idx.get) then fetch records	**O(log N + K)**
_fetch_records(movie_ids)	Convert list of IDs to movie dicts	**O(K)**
insert_movie(movie)	1) Generate new key 2) Insert into main dict 3) Insert into AVL indices	**O(log N * g)**

Graph feature explanation

GenreGraph is a graph representation of movie genres. Each genre is a vertex, and an edge exists between two genres if they appear together in the same movie. The edge weight counts the number of co-occurrences.

This allows us to analyze:

- Which genres frequently appear together.
- The most “connected” genres (high degree).
- Paths between genres (BFS shortest path).

- Aggregate metrics like total popularity for a genre.

Example queries and outputs

Operation	Query Example	Output / Description
Search by ID	qe.search_by_id(1001)	Returns the movie dictionary for ID 1001.
Search by Title	qe.search_by_title("Inception")	Returns the movie dictionary for "Inception"
Search by Year	qe.search_by_year(2015)	Returns a list of movies released in 2015.
Search by Genre	qe.search_by_genre("Action")	Returns a list of Action movies.
Insert Movie	qe.insert_movie(new_movie)	Inserts a new movie and returns its unique ID.
Modify Movie	qe.modify_movie(1336576, {"title": "The Midnight Adventure: Director's Cut", "popularity": 97.5})	Updates existing movie fields. Returns True if successful
Delete Movie	qe.delete_movie(1336576)	Deletes the movie with ID 1336576. Returns True if deletion succeeded

Summary of Findings

Total movies in database: 1336575
Years indexed: 174
Genres indexed: 19
Titles indexed: 914422

Most Common Genre: Drama (232402 movies)
Least Common Genre: Western (8878 movies)

Average Ratings (sample):

2010: 2.3214831305944457

2015: 2.3475676520750426

2020: 1.9024524407121577

Drama: 2.83

Action: 3.53

Comedy: 3.14

Sample Top Movies:

Top 3 Movies in 2020 by Rating:

1. My Senior Year (Rating: 10.0)
2. BTS 5th Muster: Magic Shop (Rating: 10.0)
3. Brother, I Cry (Rating: 10.0)